

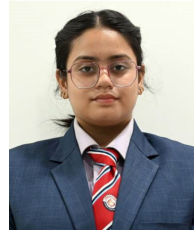
PROJECT AND TEAM INFORMATION

VeriNews is the model we propose to solve the problem.

Project Title

VeriNews: An Explainable Multi-Factor News Verification and Credibility Scoring Engine Using DSA in C/C++

Student / Team Information

Team Name: ByteForce	Team #: 07
Team member 1 (Team Lead) <i>Srivastava, Shambhavi</i> ID: 2510012471 Email: 2510012471@geu.ac.in	
Team member 2 <i>Panwar, Abhinav</i> ID: 2512510050 Email: 2512510050@geu.ac.in	
Team member 3 <i>Sharma, Akshat Mohan</i> ID: 2510010700 Email: 2510010700@geu.ac.in	

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

In today's digital environment, false news and exaggerated headlines spread quickly on social media and messaging apps. These misleading stories can change people's beliefs, create fear, and damage trust in real news. It is difficult for individuals and students to verify the truth of a headline or article as it appears due to the fast pace and large amount of unfiltered information.

Most existing fact-checking tools either depend on manual checks that are slow or use complex machine learning models that are not transparent. The real challenge is not just labeling news as fake or real, but providing users with a credibility score that gives a clear explanation of why a news item was flagged.

To address this, we are building **VeriNews**, a fast and explainable news verification engine using fundamental data structures in C and object-oriented programming in C++. By identifying key entities, matching them with verified or questionable records from curated databases, and calculating a score based on source reliability, topic relevance, and time consistency, our platform offers an easy-to-use, educational, and computationally efficient solution to help tackle misinformation from its source.

State of the Art / Current solution (1 pt)

Currently, fake news verification is mostly limited to two main approaches. One is manual fact-checking by experts, as seen in platforms like Snopes, Alt News, and PolitiFact. These are slow but accurate. On the other hand, automated systems use algorithms to detect false news quickly. However, they often lack the ability to understand context and nuances. As a result, users are forced to choose between slow but accurate methods or fast but unreliable ones. There is very little in between. Many systems also rely on cloud-based AI or NLP APIs, which are not transparent. Manual methods take too long, allowing fake news to spread before it can be corrected. On the technology front, most platforms use complex neural networks or proprietary cloud-based fact-checking tools.

These systems work like black boxes, giving users no insight into how they make decisions or how they evaluate sources. Also, existing systems rarely provide simple, standalone rule-based software that can run locally with efficient algorithms, without needing many external dependencies.

Project Goals and Milestones (2 pts)

Our goal is to build a high-performance verification and credibility-scoring application that allows users to submit news headlines or article snippets and immediately receive a weighted credibility score (0–100%) along with a detailed explanation.

Project Milestones:

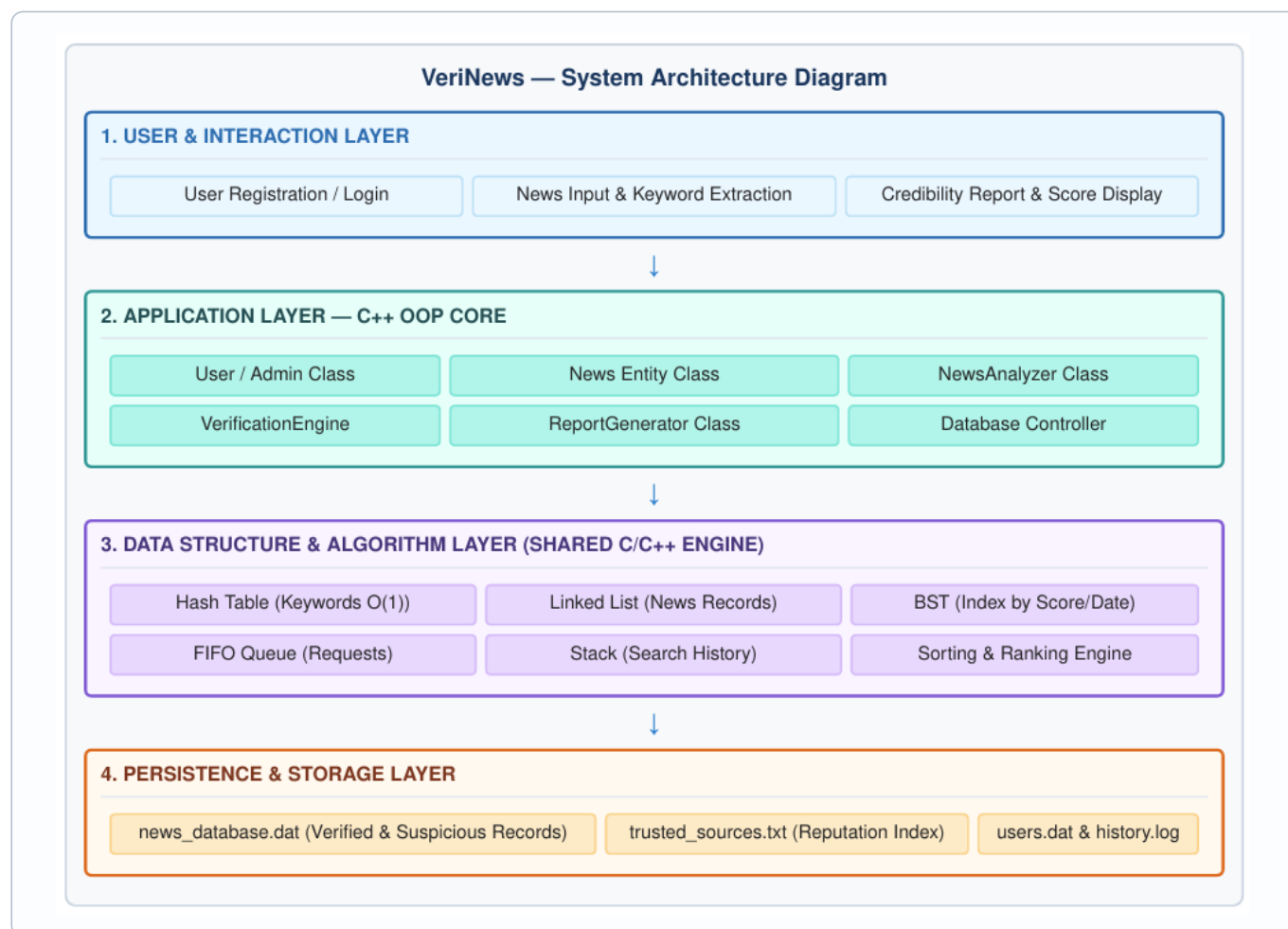
- **Milestone 1 (Phase 1):** Core Data Structure Implementation — Build C modules for Hash Tables (keyword indexing), Linked Lists (news records), and Binary Search Trees (sorting and filtering records by score and date).
- **Milestone 2 (Phase 2):** Tokenization & Verification Engine — Develop keyword extraction, source reputation scoring, and the multi-factor weighted verification algorithm.
- **Milestone 3 (Phase 2/3):** C++ OOP Layer & Request Queueing — Build class hierarchies (User, News, VerificationEngine, Admin), FIFO request queues, and undo/history stacks.
- **Milestone 4 (Phase 3):** File Persistence & Admin Module — Implement file handling for persistent news records, trusted source registries, and administrative CRUD operations.
- **Milestone 5 (Phase 3):** Testing & Comprehensive Benchmarking — Unit test matching accuracy, handle edge cases, and generate final proposal deliverables.

Project Approach (3 pts)

We are implementing this project using a modular, multi-tier architecture that combines low-level C data structures for optimal performance with high-level C++ object-oriented programming (OOP) for maintainability and modular design:

- **Text Processing & Keyword Extraction:** Clean input text, remove unnecessary elements, filter stop words, and extract key entities.
- **Data Structures & Matching Layer (C):** Hash Tables are used for quick keyword lookups in near constant time, dynamic Linked Lists are used to chain records, Binary Search Trees (BST) are used for filtering based on time or score ranges, FIFO Queues are used for grouping input data, and Stacks are used to track operation history.
- **Multi-Factor Verification Engine:** This system calculates an overall score based on several factors: Entity Match (40%), Source Reliability (25%), Corroborating Reports (20%), Temporal Consistency (10%), and Category Alignment (5%).
- **Classification & Persistence (C++ OOP):** Output is categorized into three groups: Likely Verified (80–100%), Needs Verification (50–79%), or Suspicious (0–49%).

System Architecture (High Level Diagram) (2 pts)



Project Outcome / Deliverables (1 pt)

By the end of this project, we will deliver a fully working, standalone C/C++ application called **VeriNews** that allows users to assess news headlines and view detailed credibility scores.

Key Deliverables include:

- Complete, modular, and well-documented source code that combines custom C data structures (Hash Tables, BST, Linked Lists, Queues, Stacks) with C++ OOP classes.
- A pre-configured dataset with verified news articles, flagged suspicious records, and indexes for trusted publishers.
- An interactive Administrative module that provides full Create, Read, Update, and Delete (CRUD) functionalities for managing the database and viewing audit history.
- A comprehensive project report, analysis of algorithmic complexity, UML diagrams, and a live demonstration showing real-time news verification and scoring.

Assumptions

All submissions are assumed to be in English and contain enough entities for tokenization. A baseline truth dataset is managed by administrators. The tool is designed as a fast, deterministic local verification tool, not an exhaustive real-time crawler.

References

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- [2] Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). Fake News Detection on Social Media. *ACM SIGKDD Explorations*, 19(1), 22–36. <https://doi.org/10.1145/3137597.3137600>
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